response, fleets of UAVs may need to determine the fastest rescue routes, identify safe landing zones, assess infrastructure damage, and coordinate supply delivery all simultaneously. Quantum algorithms allow drones to compute these scenarios in real time, enabling rapid adaptive decision-making under dynamic conditions. In surveillance and reconnaissance, quantum processing allows UAVs to analyze large-scale visual data streams, detect anomalies, and classify objects with high precision even in cluttered or changing environments. This capacity supports greater autonomy, efficiency, and operational safety.

9.5.3 Future applications in logistics, surveillance, and disaster response

The integration of quantum computing into drone systems holds promise across a wide range of future applications. In logistics, quantum-enabled UAVs could optimize cargo routing networks, reduce delivery times, and coordinate large autonomous fleets. In surveillance and environmental monitoring, drones could detect subtle atmospheric changes, analyze migration patterns, or monitor border and maritime zones with unprecedented analytical depth. In disaster response, quantum-capable drones could rapidly evaluate terrain conditions, locate survivors using multispectral analysis, and coordinate rescue teams through real-time swarm intelligence. These future scenarios underscore how quantum computing will enable drones to serve not only as tools of observation or delivery, but as strategic decision-support platforms across global systems.

Table 9.1 provides a comparative overview of three generations of UAV system development. Classical drones rely heavily on manual control and are limited in autonomous capability. AI-enhanced drones introduce learning-based perception, adaptive navigation, and partial autonomy, enabling more sophisticated civilian and industrial applications. Quantum-